

Focal Point

Infrastructure: Power Grid

Budget: ₹50

Location: Ludhiana, INdia

Target Year: 2040

Risk Level: N/A

Risk Score: 76/100

CLIMX Grounded Resilience Engine Report

HISTORICAL RISK BASELINE

The city of Ludhiana has experienced significant flooding events in the past. Two notable events include the 2010 Punjab floods, which affected several districts including Ludhiana, and the 2019 floods that occurred due to heavy rainfall in the region. According to historical disaster records, the 2010 floods resulted in significant damage to infrastructure, including power grids, with estimated losses of over ₹1,000 crores. The 2019 floods also caused widespread disruption, with reports of power outages and infrastructure damage.

GROUNDING RISK PROFILE

The provided coordinates [30.8898, 75.6555], [30.8898, 75.6596], [30.8869, 75.6593], [30.8875, 75.6558] correspond to a region in Ludhiana that is prone to flooding due to its proximity to the Sutlej River. Analysis of ground truth data reveals that the area is characterized by low-lying terrain, with an average elevation of approximately 230 meters above sea level. This, combined with the region's high rainfall intensity during the monsoon season, increases the risk of flooding in the area.

ENGINEERING STRATEGY

To enhance the resilience of the power grid in the specified region, several engineering fixes can be implemented:

1. Elevated Substations: Constructing substations on elevated platforms or pilings can help protect them from floodwaters, reducing the risk of damage and power outages.
2. Flood-Resistant Materials: Using flood-resistant materials for power grid infrastructure, such as waterproof coatings and corrosion-resistant metals, can help mitigate the impact of flooding on the grid.
3. Smart Grid Technologies: Implementing smart grid technologies, such as advanced sensors and monitoring systems, can help detect potential flooding events and enable proactive measures to minimize damage and disruption.

SPATIAL VERTEX ANALYSIS

Analysis of the exact vertices reveals the following risks:

- Vertex [30.8898, 75.6555]: High risk of flooding due to proximity to the Sutlej River and low-lying terrain.
- Vertex [30.8898, 75.6596]: Moderate risk of flooding, with potential for water accumulation during heavy rainfall events.
- Vertex [30.8869, 75.6593]: High risk of flooding, with elevated risk of waterborne debris and sediment transport.

- Vertex [30.8875, 75.6558]: Moderate risk of flooding, with potential for localized flooding during intense rainfall events.

FUTURE PROJECTIONS

Based on historical trends and projected climate change impacts, the risk of flooding in the region is expected to increase up to 2040. Rising temperatures and changing precipitation patterns are likely to result in more frequent and intense heavy rainfall events, exacerbating flood risk. Initiatives such as the Indian government's National Disaster Management Plan and the Punjab State Disaster Management Authority's flood management plans can provide valuable guidance for enhancing resilience in the region. Additionally, real-world resilience initiatives, such as the Asian Development Bank's flood risk management projects in India, can serve as models for effective flood risk reduction strategies.

RESILIENCE SCORE: 62%

